

Total No. of Questions : 8]

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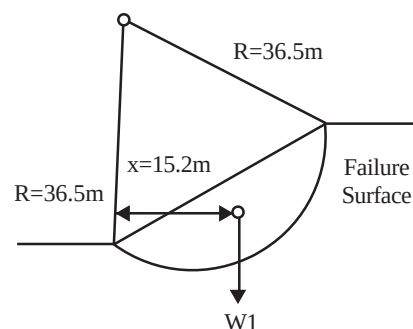
Roll No

MI-8004 (2) (CBGS)
B.E. VIII Semester Examination, June 2020
Choice Based Grading System (CBGS)
Rock Slope Engineering
Time : Three Hours

Maximum Marks : 70

Note: i) Attempt any five questions.
ii) All questions carry equal marks.

1. a) Differentiate between highwall slope and waste dump slope in terms of their characteristics from strength point of view.
b) Write notes on:
 - i) Effect of water on slope stability
 - ii) Stability of slopes and slope failure
2. a) Discuss the importance of pit slope angle on the economics of open pit mining operation.
b) Describe the method for conducting direct shear test on a rock sample containing joint to measure the shear strength available along the joint plane.
3. a) What are flow nets? Explain any one field method for measurement of permeability of rockmass.
b) Explain how will you carry out geotechnical investigations for slope stability study. What data will you collect during such investigation?
4. a) How reinforcement improves the stability of slopes? Discuss the various techniques of reinforcing slopes.
b) Compare LEM technique with FEM technique.
5. a) Derive the formula for determining the factor of safety of a slope under plane failure condition, when the tension crack lies on the surface.
b) Discuss the various geometrical conditions under which circular and wedge failure takes place.
6. a) Explain any one method based on slicing commonly used for stability analysis of circular failure. Write down the various assumptions made.
b) Explain slip circle technique of circular failure analysis. Calculate the FoS of slope against shear failure along the slip circle for the condition shown below. The soil cohesion is 40 kPa and bulk density is 2040kg/m^3 . Assume the friction angle to be zero throughout the failure surface.



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7. a) Explain under which conditions toppling failure of slopes occurs? Discuss various types of toppling failure.
b) Explain the working of slope stability radar used for monitoring slope movements.
8. What is regressive and progressive movement of slope? Explain how will you interpret the data of slope displacement for prediction of slope failure time?
